

2.2 Overview of Cloud Security

In this course, we will:

- Outline the benefits of cloud security
- Capture and locate logs
- Examine Amazon Inspector, AWS Security Hub, AWS Shield (standard and advanced), and AWS GuardDuty
- Outline encryption options at AWS
- Define governance and compliance services

Benefits of Cloud Security

Scaling Securely with High Visibility and Control



- With AWS, customers manage
 - Where their data is stored
 - Who can access their data
 - What resources the organization is using at any point in time
- Granular identity and access controls are combined with continuous monitoring for near real-time security information:
 - Guarantees that the right resources have the right access at the right times, regardless of where information is stored

Automating and Minimizing Risk with Deeply Integrated Services

- Allows enterprises to be more secure
- Can reduce human configuration errors
- Allows teams to focus on work that is business critical



- Companies can select from a variety of highly integrated solutions that can be merged to uniquely automate tasks
- The security teams and DevOps can more easily work together to create and deploy code faster and more securely



Building with the Top Standards for Privacy and Data Security

- AWS is vigilant about the privacy of all cloud customers
- Cloud customers can build on the most secure global infrastructure
- Organizations always own their data:
 - Encrypt it
 - Transport it
 - Manage the entire life cycle



Building with the Top Standards for Privacy and Data Security

- AWS automatically encrypts all data that moves over the AWS global data center and regional networks:
 - At the physical layer before leaving their secured facilities
- There are additional encryption layers over all virtual private cloud (VPC) cross-region peering traffic, and customer or service-to-service Transport Layer Security (TLS) connections

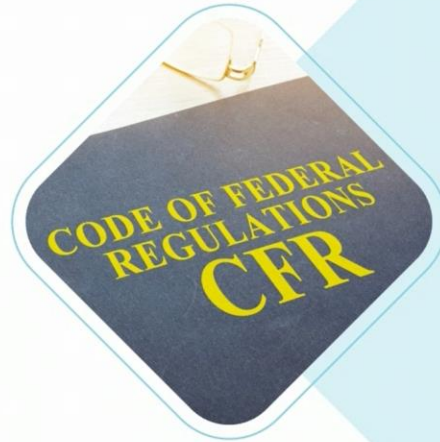
Utilizing the Biggest Ecosystem of Security Partners and Solutions

- Use security technology and consulting services from trusted solution providers like Fortinet and Palo Alto Networks
- Providers and partners with extensive expertise and proven success securing all phases of cloud adoption are selected



Inheriting the Most Extensive Security and Compliance Controls

- To assist with organizational compliance initiatives, AWS attains third-party validation for thousands of global compliance requirements
- AWS continually monitors compliance requirements to help customers meet security and compliance standards for industries like finance, retail, healthcare, and government



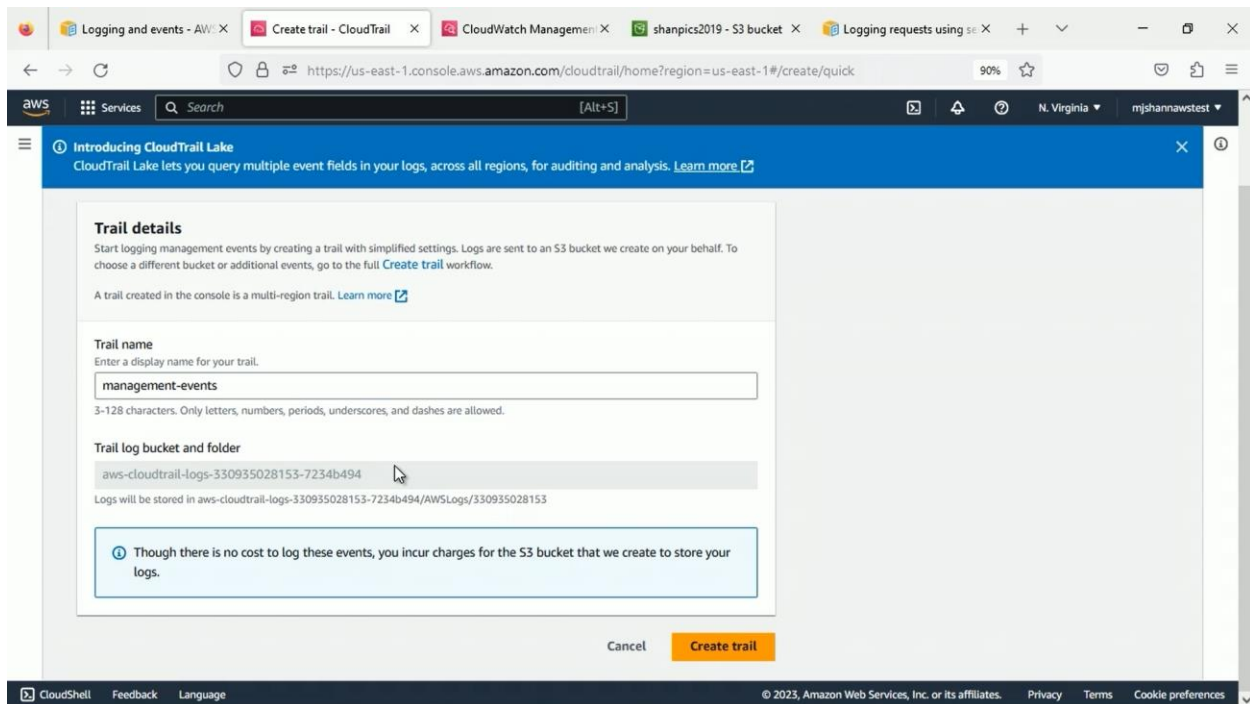
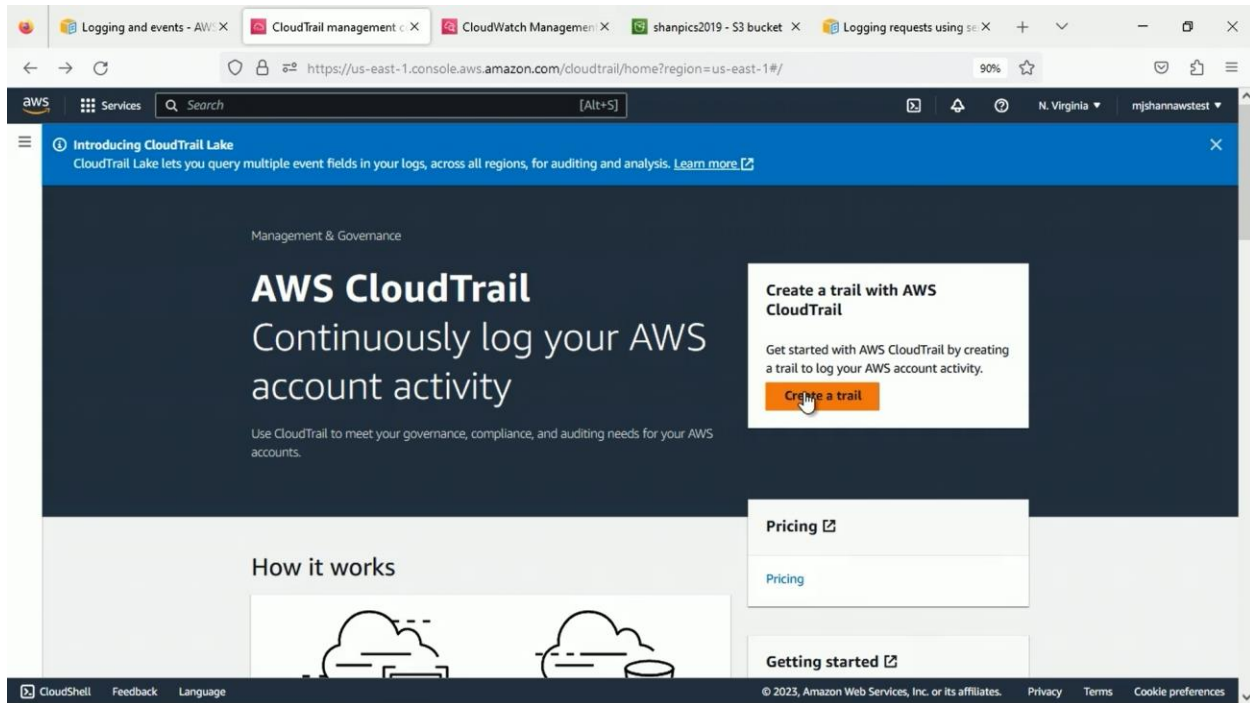
Capturing and Locating Logs

The screenshot shows a web browser window displaying the AWS Security Incident Response Guide. The browser's address bar shows the URL: <https://docs.aws.amazon.com/whitepapers/latest/aws-security-incident-response-guide/logging-and-events.html>. The page header includes the AWS logo, a search bar, and navigation links like 'Contact Us', 'English', and 'Return to the Console'. The left sidebar contains a table of contents for the 'AWS Security Incident Response Guide', with 'Logging and events' highlighted. The main content area is titled 'AWS CloudTrail' and describes the service's role in governance, compliance, and risk auditing. It lists two types of events: 'CloudTrail management events' (control plane operations) and 'CloudTrail data events' (data plane operations). Below this, it introduces 'Amazon CloudWatch Events' as a near real-time stream of system events.

AWS CloudTrail – AWS CloudTrail service enabling governance, compliance, operational auditing, and risk auditing of AWS accounts. With CloudTrail, you can log, continuously monitor, and retain account activity related to actions across AWS services. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command line tools, and other AWS services. This event history simplifies security analysis, resource change tracking, and troubleshooting. CloudTrail logs two different types of AWS API actions:

- **CloudTrail management events** (also known as control plane operations) show management operations that are performed on resources in your AWS account. This includes actions such as creating an Amazon S3 bucket and setting up logging.
- **CloudTrail data events** (also known as data plane operations) show the resource operations performed on or within a resource in your AWS account. These operations are often high-volume activities. This includes actions such as Amazon S3 object-level API activity (for example, `GetObject`, `DeleteObject`, and `PutObject` API operations) and Lambda function invocation activity.

Amazon CloudWatch Events – Amazon CloudWatch Events delivers a near real-time stream of system events that describe changes in AWS resources, or when API calls are published by AWS CloudTrail. Using simple rules that you can quickly set up, you can match events and route them to one or more target functions or streams. CloudWatch Events becomes aware of operational changes as they occur. CloudWatch Events can respond to these operational changes and take corrective action as necessary, by sending messages to respond



Logging and events - AWS X Create trail - CloudTrail X CloudWatch Management X shanpics2019 - S3 bucket X Logging requests using se X

https://docs.aws.amazon.com/whitepapers/latest/aws-security-incident-response-guide/logging-and-events.html

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AWS > Documentation > AWS Whitepapers > AWS Technical Guide Feedback Preferences

AWS Security Incident Response Guide

AWS Technical Guide

- Abstract
- Introduction
- Preparation
- Operations
- Post-incident activity
- Conclusion
- Contributors
- Appendix A: Cloud capability definitions
 - Logging and events**
 - Visibility and alerting
 - Automation
 - Secure storage

respond to these operational changes and take corrective action as necessary, by sending messages to respond to the environment, activating functions, making changes, and capturing state information. Some security services, such as Amazon GuardDuty, produce their output in the form of CloudWatch Events. Many security services also provide an option to send their outputs to Amazon S3.

AWS Config – AWS Config is a service enabling customers assess, audit, and evaluate the configurations of your AWS resources. AWS Config continuously monitors and records your AWS resource configurations and enables you to automate the evaluation of recorded configurations against desired configurations. With AWS Config, customers can review changes in configurations and relationships between AWS resources, manually or automatically, detailed resource configuration history, and determine overall compliance against the configurations specified in customer's guidelines. This enables simplification of compliance auditing, security analysis, change management, and operational troubleshooting.

Amazon S3 access logs – If sensitive information is stored in an Amazon S3 bucket, customers can enable Amazon S3 access logs to record every upload, download, and modification to that data. This log is separate from, and in addition to, the CloudTrail logs that record changes to the bucket itself (such as changing access policies and lifecycle policies). It's worth noting that access log records are delivered on a best effort basis. Most requests for a bucket that is properly configured for logging result in a delivered log record. The completeness and timeliness of server logging is not guaranteed.

Amazon CloudWatch Logs – Customers can use Amazon CloudWatch Logs to monitor, store, and access log files originating from operating systems, applications, and other sources running in Amazon EC2 instances with a CloudWatch Logs agent. CloudWatch Logs can be a destination for AWS CloudTrail, Route 53 DNS

Logging and events - AWS X Create trail - CloudTrail X CloudWatch Management X shanpics2019 - S3 bucket X Logging requests using se X

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Amazon VPC Flow Logs – VPC Flow Logs enables customers to capture information about IP traffic going to and from network interfaces in VPCs. After enabling flow logs, they can be streamed to Amazon CloudWatch Logs and Amazon S3. VPC Flow Logs helps customers with a number of tasks such as troubleshooting why specific traffic is not reaching an instance, diagnosing overly restrictive security group rules, and using it as a security tool to monitor the traffic to EC2 instances. Use the most current version of VPC flow logging to get the most robust fields.

AWS WAF Logs – AWS WAF supports full logging of all web requests inspected by the service. Customers can store these in Amazon S3 to fulfil compliance and auditing requirements, as well as debugging and forensics. These logs help customers determine root cause of initiated rules and blocked web requests. Logs can be integrated with third-party SIEM and log analysis tools.

Route 53 Resolver query logs – Route 53 Resolver query logs will let you log all DNS queries made by resources within Amazon Virtual Private Cloud (Amazon VPC). Whether it's an Amazon EC2 instance, an AWS Lambda function, or a container, if it lives in your Amazon VPC and makes a DNS query, then this feature will log it; you are then able to explore and better understand how your applications are operating.

Other AWS logs – AWS continuously releases service features and capabilities for customers with new logging and monitoring capabilities. For information about features available for each AWS service, refer to our public documentation.

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/resolver-query-logs.html help you? Next topic: Visibility and alerting

Amazon Inspector

Amazon Inspector

- Is a managed vulnerability assessment service that continuously scans AWS workloads for software vulnerabilities, gaps, and unintentional network exposures
- Automatically finds and scans running Elastic Compute Cloud (EC2) instances, container images in Amazon Elastic Container Registry (ECR) and AWS Lambda functions for known vulnerabilities and unintended exposure

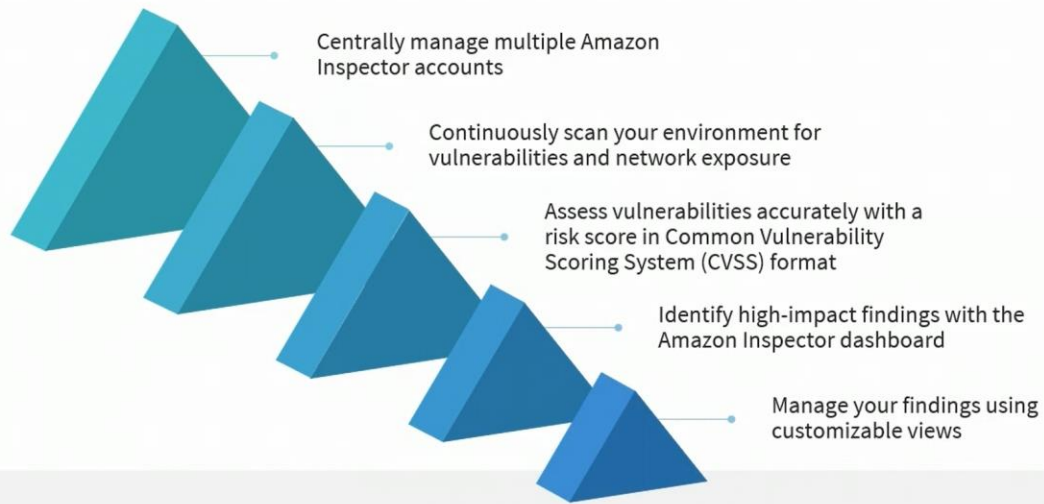


Amazon Inspector



- Amazon Inspector creates a finding when it discovers a software vulnerability or network configuration issue
- A finding describes the vulnerability, identifies the affected resource, rates the severity of the vulnerability, and provides remediation guidance
- Customers can analyze results using the Amazon Inspector console or visualize and process discoveries through other associated AWS services

Amazon Inspector



Using AWS Security Hub

Security Hub | us-east-1

CloudWatch Management Console

https://us-east-1.console.aws.amazon.com/securityhub/home?region=us-east-1#/landing

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use AWS System Manager Automation documents, AWS Step Functions, and AWS Lambda functions to build automated remediation workflows that can be initiated from Security Hub.

while individual account owners see only findings associated with their account but across all the Regions for that account. Security Hub is also integrated with AWS Organizations to automatically enable new accounts in your organization with Security Hub.

Use Cases

Security scanning Continuously scan your AWS accounts for configuration errors using various security standards and roll up security check results at the account and multi-account level to understand your overall security state.	Triage and prioritize security issues Use AWS Security Hub's summary dashboards and filtering rules to identify and prioritize which findings from AWS security services and partner security integrations are most important and which require the most immediate attention.
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Related services

Amazon GuardDuty Threat detection and continuous	Amazon Inspector Analyze the behavior of your AWS
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CloudShell Feedback Language

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Security Hub | us-east-1 CloudWatch Management Console

https://us-east-1.console.aws.amazon.com/securityhub/home?region=us-east-1#/landing

How it works

Automated security checks
Run automatic security checks across your AWS environment

Consolidated, prioritized security findings
Amazon GuardDuty
Amazon Inspector
Amazon Macie
+ Additional AWS services and integrated partner solutions

Take action
Build customized actions and send findings to ticketing, chat, email, or automated remediation systems using integration with Amazon EventBridge.

Security checks
Per account, per month, per Region

First 100,000	\$0.0010/check
100,001 – 500,000	\$0.0008/check
500,001 +	\$0.0005/check

Finding ingestion events
Includes ingestion of updates to existing findings. Finding ingestions for Security Hub security checks are free.

First 10,000	Free
10,001 +	\$0.00003/finding

More resources

- [What is AWS Security Hub?](#)
- [API Reference](#)
- [FAQs](#)

Benefits and features

CloudShell Feedback Language

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Security Hub | us-east-1 CloudWatch Management Console

https://us-east-1.console.aws.amazon.com/securityhub/home?region=us-east-1#/onboard

Before you can enable Security Hub standards and controls, you must first enable resource recording in AWS Config. You must enable resource recording for all of the accounts and in all of the Regions where you plan to enable Security Hub standards and controls. If you do not first enable resource recording, you might experience problems when you enable Security Hub standards and controls. AWS Config bills separately for resource recording. For details, see the [AWS Config pricing page](#).

You can enable resource recording manually from the [AWS Config console](#), or you can choose Download to download and then deploy an AWS CloudFormation template as a StackSet. See our [documentation](#) for more details.

[Download](#)

Security standards

Enabling AWS Security Hub grants it permissions to conduct security checks. [Service Linked Roles \(SLRs\)](#) with the following services are used to conduct security checks: Amazon CloudWatch, Amazon SNS, AWS Config, and AWS CloudTrail.

- ☒ Enable AWS Foundational Security Best Practices v1.0.0
- ☒ Enable CIS AWS Foundations Benchmark v1.2.0
- ☐ Enable CIS AWS Foundations Benchmark v1.4.0
- ☐ Enable NIST Special Publication 800-53 Revision 5
- ☐ Enable PCI DSS v3.2.1

AWS Integrations

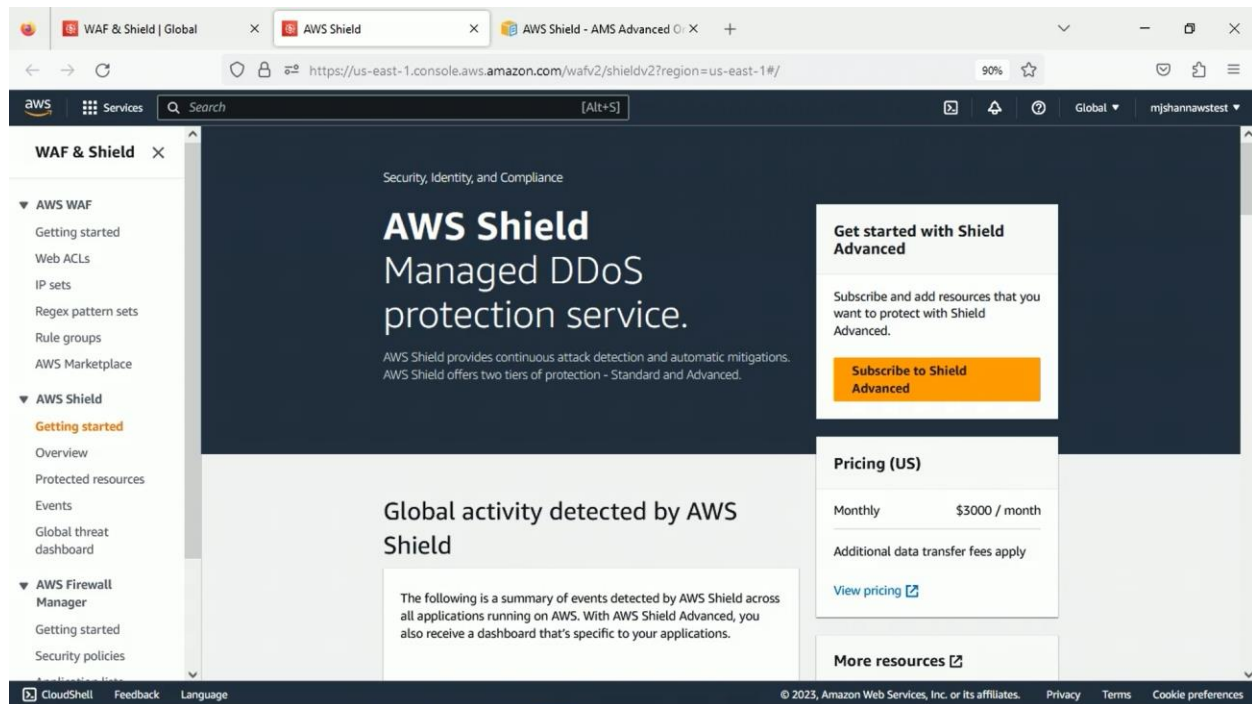
Enabling Security Hub grants it permissions to import findings from AWS services that you have enabled.

[Learn more](#)

CloudShell Feedback Language

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Using AWS Shield



AWS GuardDuty

AWS GuardDuty



- Is a managed threat detection service that constantly monitors for malevolent and unauthorized behavior
- Watches out for unusual application programming interface (API) calls or potentially unauthorized deployments that indicate a possible account compromise (zero day)
- Detects potentially compromised instances or reconnaissance by advanced persistent threat actors

-



- The finding details include information about what occurred, what AWS resources were involved in the suspicious activity, when this activity took place, and other data
- The finding type provides a description of the potential security issue:
Recon:EC2/PortProbeUnprotectedPort
- GuardDuty now uses AI to conduct “pre-crime” identification of bad actors 7-10 days in advance of malware release

Comparing Encryption Options at AWS

- EBS
- S3

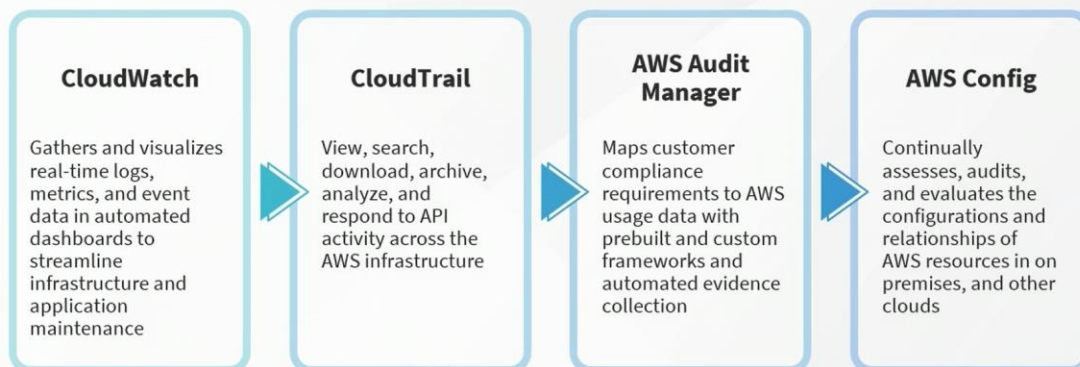
Governance and Compliance Services

Governance and Compliance Services

- There are a plethora of services that assist in governance and compliance at Amazon Web Services
- Although the following services have a number of different use cases, Amazon CloudWatch, AWS CloudTrail, AWS Audit Manager, and AWS Config are commonly utilized to support organizational governance and compliance with access reporting



Governance and Compliance Services



1. Which service offers customers a comprehensive view of their security state against security industry standards and best practices such as the Center for Internet Security (CIS)?
 - **Security Hub**

2. Which of these are key services that are commonly utilized to support organizational governance and compliance with access reporting?
 - **AWS CloudTrail**
 - **AWS Config**
 - **AWS Audit Manager**
 - **Amazon CloudWatch**

3. When using AWS security, customers benefit from using security technology and consulting services from trusted solution providers like Fortinet and Palo Alto Networks. Which AWS security propositions is this related to?
 - **Utilizing the biggest ecosystem of security partners and solutions**

4. Which is an always-free anti-DDoS service that runs constantly to protect all accounts in good standing?
 - **Shield Standard**

5. Which managed security service lets customers create, manage, and control cryptographic keys across their applications and AWS services?
 - **AWS Key Management Service**

6. Which fully managed security service uses “findings” to look for unusual API calls or potentially unauthorized deployments that indicate a possible account compromise (zero day)?
 - **AWS GuardDuty**

7. What is a critical security API capture solution that enables operational and risk auditing, governance, and compliance of your AWS account?
 - **CloudTrail**

8. Which is a managed vulnerability assessment service that continuously scans AWS workloads for software vulnerabilities, gaps, and unintentional network exposures?
 - **Amazon Inspector**